Assessing the Risk Factors of Under-Five Child Mortality: A Study Based on BDHS 2022

Submitted by

Abu Sufiun Rhyme

Roll No: FH-182-003

Session: 2023-24

Supervised by

Md. Mynul Islam

Assistant Professor, ISRT

A Project Submitted for Partial Fulfillment of the Requirement for the Degree of B.S. in Applied Statistics

Institute of Statistical Research and Training

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Declaration

I, hereby, certify that this project report entitled *"Assessing the Risk Factors of Under-Five Child Mortality: A Study Based on BDHS 2022"* submitted as a partial requirement for the degree of B.S. in Applied Statistics is the result of my own research. This study in whole or in part has not been submitted for an award, including higher degree to any other University or Institution.

Name: Abu Sufiun Rhyme

Signature:....................

Date:...........................

Acknowledgments

At first I want to express my gratitude to the almighty Allah, for giving me strength, knowledge and courage to complete this research.

I express my deepest gratitude to my respected supervisor, **Md. Mynul Islam** , Assistant Professor, Institute of Statistical Research and Training (I.S.R.T.), University of Dhaka, whose great effort guided me throughout my study. This research would not have been possible without his continuous guidance and support.

Furthermore, I would like to express my heartfelt thanks to all of my teachers and friends for helping and inspiring me. And last but not the least, I wish to thank my family members specially my parents for their encouragement and support.

Dedicated

to

My Family

Abstract

Child mortality is pivotal in countries like Bangladesh, especially considering its status as a low- to middle-income nation. This study investigated the influencing factor of Under-Five Child Mortality utilization in Bangladesh using data from the 2022 Bangladesh Demographic and Health Survey through multilevel logistics regression. Exploratory analysis has been used to speculate the distribution of the explanatory variables and under-five child mortality. Bivariate analysis has been performed to find out the variation in the prevalence of under-five child mortality with different background factors. Multilevel logistic regression has been conducted to identify the associated background factors. The analysis incorporates significant factors such as mother's education, father's education, sex of children, birth weight, breastfeeding, mode of delivery, and division using a multilevel logistic regression model. An increase in father's and mother's education is associated with a decrease in child mortality. Normal birth weight reduces child mortality by 90% compared to low birth weight. Additionally, C-section deliveries are associated with a 70% reduction in mortality compared to vaginal deliveries. The highest probability of child mortality is in mymensingh. This study shows that child mortality varies by the sex of the child. Both the government and individuals should place greater emphasis on addressing these factors to help reduce under-five child mortality.

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Chapter 1

Introduction

1.1 Background of study

Child mortality, also known as mortality under five years of age, refers to the probability that a child born in a specific year will die before reaching the age of five, expressed per 1,000 live births (World Health Organization, 2023). The mortality rate for children or children under five years of age is a crucial indicator to navigate the early health status of a child (UNICEF, 2023). In addition, it reflects the effectiveness of the health system, the public health intervention, and the broader socioeconomic conditions of the child cohort of a region.

Increment or reduction in child mortality is somewhat directly stipulated to the SDG fundamental goals, especially the third goal: Ensure healthy living and promote well-being for all ages. This goal explicitly targets the reduction of maternal mortality (Target 3.1), ending preventable deaths of newborns and children under five years of age (Target 3.2), and combating communicable and non-communicable diseases (United Nations, 2015).

According to the report until 2023, the global mortality rate for under five years of age has dropped by 59% since 1990, falling from 93 to 37 deaths per 1,000 live births. Nevertheless, 4.8 million children under the age of five died in 2023, with approximately half (47%) of these deaths occurring in the neonatal period (the first 28 days of life). Regional disparities has a role to play i.e sub-Saharan Africa and Southern Asia accounting for more than 80% of under-five deaths. Pneumonia, diarrhea, malaria, preterm birth complication, and birth-related problems are the precursors of these deaths (World Health Organization, 2023).

In Bangladesh, noticeable progress has been made over the decades to reduce mortality, especially among children under five years of age, thanks to expanded immunization coverage, improved maternal healthcare, and greater access to basic health services. According to the Bangladesh Demographic and Health Survey (BDHS), the mortality rate of children under five years of age has shown significant declination, dropping from 143.8 in 1990 to 31 in 2022 of 1000 newborns BDHS (1993, 2022).

However, it is the call of an hour that the confluence of public health intervention influencing the child mortality and it's associated factors get reviewed over the time in order to keep standing on the track of lowering the child mortality rate. Economic growth and socio-economic status is a fluid factor that varies over the time thus impacting the correspondent child mortality rate. Thereby continuation of study with fresh information is necessary to decipher the analogous determinants affecting child mortality (You et al., 2015).

1.2 Literature Review

Child mortality is one of most critically studied topic across the public health domain, which is used to describe the deaths of children under five. Navigating the various aspects of child mortality, along with it's causes and trends over the time, and the outcomes of programs and measures aimed at lowering it, this literature review pinpoints the landscape of this phenomenon.

Deciphering the influence of socio-demographic factors on child mortality in regional disparities using cross-sectional survey data was the major goal of this study(Haq et al., 2020). Factors like preceding birth interval, women's status, and others, had significant influence and with the upgrade in education level, the risk was decreased.

Usage of contemporary contraceptives in Bangladesh has been a key for addressing child mortality issue, for instance women whose children are under five have a greater chance of surviving if the are affirmative about the usage of contraception. Education level, age, and other preexisting variables was found to be effective as usual. The Weibull model yields more effective result for this particular study. In addition, it highlights how crucial it is to utilize contraceptives in Bangladesh (Khan et al., 2023).

Mother's age at birth (< 20), and specific division (i.e Sylhet) was found to be significant in this study. The Cox proportional hazard model was used using the Bangladesh Demographic and Health Survey (BDHS) 2017-18 to identify the socioeconomic and demographic risk factors for under-five mortality in Bangladesh (Soni et al., 2023).

Using the MICS 2019, a study explored that Bangladesh had a mortality rate for under five age of children of 33.3 per 1000 live births. The women's and children's databases were used in this investigation. Prominent risk factors for under-five child mortality was explored using the bivariate analysis (Ara et al., 2023).

Socio-demographic determinants along with cultural traits affects the mortality rates. For instance, early marriage as a cultural trait has a role to play, thus found to be significant in this study. This study also deciphers the morbidity rate and trend analysis of the mortality indicators from 1993 to 2018 (Hossain et al., 2022).

Child-mother interaction based variables such as sex of the child, birth order, mother's age at birth were also figured out to be significant in several studies. This particular study indicates that the birth spacing focused public health initiatives and community based educational initiatives may prove to be the most successful (Khan and Awan, 2017).

Several studies explored the regional disparities on micro level i.e district level and temporal region. This study digged in distinguishing the differences in mortality rate across the districts over the time period. Complete cases from the Sample Vital Registration System (SVRS) were among the seven surveys whose data was used. A linear mixed effects model (LMM) was used to showcase the negative correlation between IMR and life expectancy. This study also recommended initiatives for mothers residing in high-risk areas (Islam and Akter, 2023).

Serendipitous calamities such as COVID-19 had a role to play in shaping the overall health intervention and public health landscape. The overall economic growth, despite the inclination in GDPs, were severely damaged by the COVID-19 pandemic. Foreign remittance which is somewhat the most crucial contributor for the economic growth for the low-income countries were severely hampered due to the lockdown, and the general state of the economy has also declined, with particular effects on mother and child health. This study (Ahmmed et al., 2021) explored the effect of COVID-19 on the financial circumstances of Bangladesh.

Many studies jumped into expanding the gender disparities of the mortality rates and elongated explaining the associated factors responsible behind. This particular study concluded that household level factors such as quintile of wealth, household size, and hand-washing materials were discovered to have a significant effect on mortality. In addition, this study suggests that boys are more likely to have morbidity than that of girls (Methun et al., 2021).

Pre-birth complication and histories are hard to neglect alike factors while explaining under five mortality. Several studies are available which explicitly explained factors like preterm birth complications, pneumonia, and birth asphyxia as cause of death. This study also recommended targeted interventions to prevent cause specific death (Mazumder et al., 2024).

Advanced analytical approaches such as machine learning techniques were also used to analyse and decompose factors associated with under five mortality rates. Availability of healthcare services was found to be significant in this particular study. This study (Rahman and Rahman, 2025) also paved the way for complex and deeper insight for the further quincuncial studies.

Similarly antenatal care provider skill, breastfeeding immediately after delivery, mom age, education, gender and previous birth interval affects the infant mortality rates in Bangladesh. Statistically speaking, 28 live-birth out of 1000, Bangladesh has a high infant mortality rate. The chance of survival rises with early care. This study also emphasizes on policy recommendations to lower infant mortality and reach the sustainable development goals (Jamee et al., 2022).

These studies summarized above critically navigate the existing literature on under five mortality and provide a thorough detail of the different factors that contribute to child mortality in Bangladesh. They also pinpointed the overall catalogs that affects the mortality rates such as child level factors, maternal and paternal determinants along with household and community level indicators. Furthermore, they paved the way to initiate new studies and necessity of policy recommendations.

1.3 Objective of the study

This study aims to determine the prevalence of child mortality and systematically examine its underlying determinants in the context of Bangladesh, using data Bangladesh Demographic and Health Survey (BDHS) 2022. The research seeks to contribute meaningfully to the country's public health efforts by providing evidence-based insights into the factors influencing child mortality. It also supports progress toward international development targets, including Millennium Development Goal 4 (reducing child mortality) and Sustainable Development Goal 3 (ensuring healthy lives and promoting well-being at all ages, particularly through reductions in neonatal, under-five, and maternal mortality, as well as overall death rates).

Chapter 2

Data and variables

2.1 Data source and sampling design

This study utilized secondary data from the Bangladesh Demographic and Health Survey (BDHS) 2022, which is a part of Demographic and Health Survey (DHS) program. The Demographic and Health Survey (DHS) is a globally recognized program which provides nationally representative household survey. The DHS survey collects various types of data such as household, health, and nutrition indicators. This survey provides valuable insights that help to develop policies, programs, and national development goals. It also supports tracking the progress of the Sustainable Development Goals (SDGs) in our country. The first Demographic and Health Survey (DHS) program was conducted in Peru in 1984. Since then, the program has covered more than 90 countries.

The Bangladesh Demographic and Health Survey (BDHS) 2022 is the ninth installment nationally representative survey, carried out from 27 June to 12 December 2022. This survey was conducted by the National Institute of Population Research and Training (NIPORT) within the Medical Education and Family Welfare Division of the Ministry of Health and Family Welfare. Mitra and Associates, a private research agency of Bangladesh, implemented the fieldwork and collected data for the BDHS 2022. The survey was funded by the United States Agency for International Development (USAID)/Bangladesh. ICF provided the technical and logistical support to implement the BHDS 2022.

The sampling frame is a stratified two-stage cluster sampling. In the first stage, 675 Enumeration Areas (EAs) were selected from 293,000 EAs using a probability proportional to the size of the EAs. The Bangladesh Bureau of Statistics (BBS) was prepared these Enumeration Area based on Population and Housing Census 2011 of the People's Republic of Bangladesh. These Enumeration

Areas (EAs) were selected from eight division which were divided into two main strata : urban and rural areas. Out of the selected 675 Enumeration Areas (EAs) , 237 were from urban areas and 438 from rurals. In the second stage , 45 households ware chosen from each EAs using by systematic sampling method. In 30 of the selected 45 households interviews were using the long Questionnaire (including detailed questions on background characteristics and reproductive history) of eligible 15-49 aged woman. In the remaining 15 households were interviewed using the short Questionnaire. Total of 30,375 residential households were selected (19,710 from rural areas and 10,665 from urban areas) and 30,018 households were successfully interviewed. Out of the interviewed household , 30,358 ever-married women age 15–49 were eligible for individual interviews.

The BDHS 2022 collect various types of information to assess the demographic and health condition. Collected Demographics and Socio-economics (including age, education, marital status wealth index, Migration, and living arrangements etc.), Nutrition and Biomarkers (including Height and weight of women and children, Blood pressure, blood glucose, and Body Mass Index (BMI) etc.), household-level Information (including housing materials, water, sanitation, ownership of TV, radio, and mobile phones, and access to electricity and clean cooking fuel etc.) , gender and social issues (including domestic violence, Women's empowerment, decision-making, attitudes toward wife-beating, gender equality and so on.), women and family health (including fertility, birth history , mortality, ANC, PNC, childhood illnesses and treatments data, breastfeeding and infant feeding practices data, and so on.), mental health (including anxiety and depression data), and monitoring and evaluation (including national health programs and interventions related data, and key indicator of Sustainable Development Goals (SDGs)) and so on (BDHS, 2022).

2.2 Variable description

There are two are two types of variables in this study:

- Dependent variable / response variable
- Independent variable / explanatory variable / covariate

2.2.1 Dependent variable

In this study , there is one dependent variable (or response variable) . The response variable is "child mortality status". This variable has two categories: dead and not dead. It takes the value 1 if the child is death at or before age 5 and 0 if the child is not death at or before age 5.

2.2.2 Independent variables

There are several variables were selected as a explanatory variables in this study. These explanatory variable were selected based on literature review. The risk factors identified in previous research provide a valuable framework to assess the cause of under 5 child mortality. These variables were divided into three categories such as children Attributes, household Attributes, and parental Attributes.

Children Attributes

Table 2.1: Description of children Attributes

Variable	**Description**	**Coding**
`Sex of child`	Male	1
	Female	2
`Birth order`	One	1
	Two	2
	Three or more	3
`Birth Weight`	Low birth weight: less than 2500 gm	1
	Normal birth weight : 2500-4000 gm	2
	Over birth weight : greater than 4000 gm	3
`Breastfeeding`	Yes	1
	No	2
`Preceding birth interval`	Short: $8 - 24$ month	1
	Normal: 24+ month	2
`Mode of delivery`	Caesarean section	1
	Vaginal	2
`Place of delivery`	Home	1
	Public Hospital	2
	Private Hospital	3
`Birth attendant during delivery`	Trained	1
	untrained	2

Parental Attributes

Table 2.2: Description of parental Attributes

Variable	Description	Coding
Mother's Education	No education	1
	Primary	2
	Secondary	3
	Higher	4
Father's Education	No education	1
	Primary	2
	Secondary	3
	Higher	4
Mother's Occupation	Working	1
	Not working	2
Mother age at child birth	<20 year	1
	$20 - 30$ year	2
	> 30 year	3
Antenatal Care (ANC)	< 4	1
	$>= 4$	2
Media exposure	Yes (at least once a week watching tv, listening radio, or reading newspaper)	1
	No (else)	2

Household Attributes

Table 2.3: Description of household Attributes

Variable	Description	Coding
Division	Barishal	1
	Chattogram	2
	Dhaka	3

Table 2.3 continued from previous page

Variable	Description	Coding
	Khulna	4
	Mymensingh	5
	Rajshahi	6
	Rangpur	7
	Sylhet	8
`Residence`	Urban	1
	Rural	2
`Wealth index`	Poorest	1
	Poor	2
	Middle	3
	Rich	4
	Richest	5

2.3 Data Extraction

As already mentioned, this study utilized secondary data from the Bangladesh Demographic and Health Survey (BDHS) 2022. We extracted data on Sex of child, Birth weight, Birth order, Breastfeeding, Mother's Education, Father's Education, Mother's Occupation, Mother age at child birth, Antenatal Care (ANC), Preceding birth interval, Birth attendant during delivery, Place of delivery, Media exposure, Division,Residence, Wealth index, and child dead status for children aged 5 years or younger from BDHS 2022.

Chapter 3

Methodology

3.1 Introduction

The full theoretical framework of this research is discussed in this chapter. The study focuses on identifying the factors that influence under-five child mortality in Bangladesh. Initially, a univariate analysis is employed to determine the prevalence of the each explanatory variable and the response variable. Then, bivariate analysis is used to get a preliminary idea of the association between the response and the explanatory variables. It gives us an unadjusted association of the each explanatory variable with the response variable. However, to obtain an adjusted association of the each explanatory variable with the response variable simultaneously, a multivariate logistic regression analysis is required. The sampling method of BDHS-2022 is two-stage cluster sampling and our response variable is binary. Therefore, the multilevel logistic regression is appropriate to assess the relationship between response and explanatory variables. So, we explore several bivariate analysis methods such as Pearson's chi-square (χ^2) test and logistic regression. In addition, we discuss the concepts of the significance level, p-value and adjusted odds ratio.

3.2 Univariate Analysis

A univariate analysis analyzes each variable separately. It provides the distributions of the each explanatory variable and the response variable without adjusting for other variable and without considering any relationship among them. The steps including univariate analysis determine frequency and proportion of the each explanatory variable and the response variable individually. Also determine prevalences for each level of the each explanatory variable and the response variable.

3.3 Bivariate Analysis

Bivariate analysis is a statistical method that investigates the relationship between two variable concurrently. Bivariate analysis explores the relationship between two variable without considering the influence of other variables. It also evaluates the strength of the association between two variable. A bivariate analysis can be conduct in several way such as:

- **Two Qualitative variable:** When two variable are qualitative or categorical, a contingency table can be constructed and perform Pearson's Chi-square (χ^2) Test of Independence.
- **Two Quantitative variable:** The Pearson correlation coefficient measure the direction and strength of linear relationship of two quantitative or numerical variable. The range of Pearson correlation coefficient from -1 to +1. "-1" indicates a perfect negative linear relationship between two variables. "+1" indicates a perfect positive linear relationship between two variables. "0" indicates that there are no linear association bewteen two numerical variables.
- **Mixed type variables:** By creating a cross-tabulation of both numerical and categorical variable, it shows the distributions of numerical variable across the categorical variable.

3.3.1 Pearson's Chi-square (χ^2) Test of Independence

In the early 1900s, British statistician Karl Pearson derived Chi-square (χ^2) test of independence to investigate the association between two qualitative or categorical variables. This test based on the comparison of the observed frequencies and the expected frequencies assumed under the null hypothesis within a contingency table(Pearson, 1900). The hypotheses of chi-square test to test whether two variables are independent or significantly associated:

H_0: There is no significant association between the two categorical variables (independence).
H_1: There is a significant association between the two categorical variables (dependence).

The contingency table of two type variable such as independent variables and dependent variables is $X_1, X_2, \ldots, X_c$ and $Y_1, Y_2, \ldots, Y_r$ respectively. The observed frequencies from contingency table is denoted by O_ij, where i and j represents the row and column number of the contingency table.

The ith row total is denoted by R_i, where:

$$R_i = \sum_{j=1}^{c} O_{ij}.$$

The jth column total is denoted by C_j, where:

$$C_j = \sum_{i=1}^{r} O_{ij}.$$

The total observation is denoted by N, where:

$$N = \sum_{i=1}^{r}\sum_{j=1}^{c} O_{ij}.$$

Table 3.1: Observed contingency table

Row variables	Column Variables				Row total
	X_1	X_2	$\ldots$	X_c	
Y_1	O_{11}	O_{12}	$\ldots$	O_{1c}	R_1
Y_2	O_{21}	O_{22}	$\ldots$	O_{2c}	R_2
$\vdots$		$\vdots$	$\ddots$	$\vdots$	$\vdots$
Y_r	O_{r1}	O_{r2}	$\ldots$	O_{rc}	R_r
Column total	C_1	C_2	$\ldots$	C_c	N

The Expected frequencies denoted by E_{ij} which means the expected number of count for ith row and jth column assumed under the null hypothesis (i.e., two variable are independent).
For each cell, the expected frequency:

$$E_{ij} = \frac{R_i \times C_j}{N}.$$

The Chi-square statistics is given by,

$$\chi^2 = \sum_{i=1}^{r}\sum_{j=1}^{c} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}.$$

the χ^2 statistics follow the chi-square distribution under the null hypothesis (i.e., that two variables are independent) with $df = (r-1) \times (c-1)$ degrees of freedom. To assess the association, the Chi-square statistics (Eq.3.1) is used. It provides the p-value. A predetermined significance level (α) is chosen. If the P-value is less than or equal to the chosen significance level (α), the null hypothesis (H_0) is rejected. Here, P-value is derive from chi-square distribution. Comparing chi-square statistic's value and critical value is another method to draw conclusions. In this context, critical value obtained from chi-square distribution with $(r-1)\times(c-1)$ degrees of freedom, where r is the number of rows and c is the number of columns. If computed χ^2 is greater than critical value, we can reject the null hypothesis (H_0) (Howell, 2011). The Chi-square test of independence only assesses the association between two variables. In other words, it determines whether the variables are related or not, but it does not measure the strength or degree of their association. (Rana and Singhal, 2015).

3.4 Logistic Regression Model

3.4.1 Binary Logistic Regression Model

The bivariate analysis discussed in the previous section to assess the association between two variables, but it cannot adjust potential confounding factors. To account for these confounders, we can use a multivariate technique.

A simple binary logistic regression, is a method that used to assess the relationship between a binary response variable (i.e., that outcomes yes / no, success / failure) and an explanatory variable while accounting the potential influence of confounding factors. Let Y be binary response variable (i.e., that otcomes 0/1) and X be an explanatory variable. Binary logistic regression provides the conditonal probability of occurring $Y = 1$ given $X = x$ which are then converted into log odds and odds. The conditional probability of Y given X is denoted by $\pi(X)$. Then

$$\pi(X) = Pr(Y = 1|X = x).$$

The range of the conditional probability of Y given X ($\pi(X)$) is 0 to 1 but the systemic part of regression provides values in the range$(-\infty, +\infty)$. So, we model the logit transformation as:

$$logit(\pi(X)) = log(\frac{\pi(X)}{1 - \pi(X)}).$$

The simple binary logistic regression model is :

$$logit(\pi(X)) = log(\frac{\pi(X)}{1 - \pi(X)}) = \beta_0 + \beta_1 X,$$

where β_0 and β_1 are model parameters. β_0 represents the log odds of the event occurring when the explanatory variable is equal to zero and $\beta_1$represents the increase in the log odds of the event occurring when the explanatory variable is one-unit increase.

When multiple explanatory variables are present, the simple logistic regression model becomes multiple logistic regression model. Suppose a $p-$ dimensional vector explanatory variables as $X = (X_1, X_2, \ldots, X_p)$, which are independent. The observed values of $X = (X_1, X_2, \ldots, X_p)$ is $x = (x_1, x_2, \ldots, x_p)$. Then, the multiple logistic regression model is:

$$logit(\pi(X)) = log(\frac{\pi(X)}{1 - \pi(X)}) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p,$$

where $\beta_i, i = 1, 2, \ldots, p$ represents the parameters of the model.

3.4.2 Multilevel Logistic Regression Model

This study utilized the Bangladesh Demographic and Health Survey (BDHS) 2022, which sampling method is a two-stage cluster sampling method. So, the women sampled from the same cluster within a division exhibit similar characteristics. As a result, responses within clusters are correlated, which cannot be accurately captured by a fixed-effects logistic regression model. Therefore, a fixed-effects logistic regression model may provides a inaccurate estimates. To overcome this limitation, a multilevel (mixed-effects) logistic regression model will be utilized.

When the data has a hierarchical structure and the response variable is binary, a multilevel logistic regression model an effective analytical approach. This model accounts for the varying log-odds of the outcome being 1 instead of 0 across different clusters. To select the best multilevel logistic regression model, we can follow a three-step method proposed by (Sommet and Morselli, 2017).

Let, Y_{ij} denote the response of the ith individual in the jth cluster. For the same individual, $X_{ij} = (X_{1ij}, X_{2ij}, \ldots, X_{pij})$ denote the vector of explanatory variables, and corresponding observed values denoted by $x_{ij} = (x_{1ij}, x_{2ij}, \ldots, x_{pij})$. The conditional probability of the response variable the ith individual in the jth cluster is equal to 1 instead of 0 is denoted by π_{ij}.

We employ grand mean centering or cluster-mean centering to the lower-level predictors, depending on the type of effects we want to estimate. At first, we calculate the log odds of the outcome being 1 instead of 0 vary among the clusters. To capture the variation of the cluster, we can used only intercept term model. This model structure is given by:

$$logit(\pi_{ij}) = log(\frac{\pi_{ij}}{1 - \pi_{ij}}) = \beta_{00} + u_{0j},$$

Where, β_{00} is the fixed intercept and u_{0j} is the difference between the fixed intercept and intercept of the jth cluster. Then we will calculate the Intraclass Correlation Coefficient (ICC). The Intraclass Correlation Coefficient (ICC) provides the measure the how much the total variation of response variable (Y) is due to differences among clusters, given that the variance of the standard logistic distribution is $\frac{\pi_2}{3}$. The formula of the Intraclass Correlation Coefficient (ICC) is:

$$ICC = \frac{u_{0j}}{u_{0j} + \frac{\pi_2}{3}}.$$

Similar to the intercept, the slope parameter of a predictors can be vary by cluster. To choose whether to use a fixed slope parameter or a random slope parameter of a predictors , we can use likelihood ratio test. The last step is to fit the final model with the chosen fixed or random intercept and slope terms.

The formulation of a multilevel multiple logistic regression model incorporating random intercepts can be described as follows:

$$logit(\pi_{ij}) = log(\frac{\pi_{ij}}{1-\pi_{ij}}) = (\beta_{00} + u_{0j}) + \beta_{01}x_{1ij} + \beta_{02}x_{2ij} + ... + \beta_{0p}x_{pij}, \tag{3.1}$$

where β_{oj}'s in (3.1) represent as the parameters of the multilevel multiple logistic regression model for $j = 0, 1, \ldots, j$ (Alam et al., 2024).

3.4.3 Odds Ratio

The quantity $\frac{\pi(x)}{1-\pi(x)}$ is called odds and $log\frac{\pi(x)}{1-\pi(x)}$ is called log odds. The ratio is defined as the ratio of the odds for $Y = 1$ to the odds for $Y = 0$. This ratio expresses the likelihood of Y belonging to a particular category compared to another. The effect of the independent variable on the outcome variable can be assessed through the odds ratio after the necessary parameters have been estimated. The odds ratio provides a value ranging from 0 to infinity. The interpretation of the odds ratio (**OR**) is as following:

- **OR** $>$ **1:** This indicates that the odds of an event are higher in one group compare to the other.
- **OR** $<$ **1:** This indicates that the odds of an event are lower in one group compare to the other.
- **OR** $=$ **1:** This indicates that there is no difference in the odds between the two groups.

Adjustment of odds ratio is needed because other factors can have impact on the relationship between explanatory and response variable. These factors that have influence are known as confounders, so after balancing them in the analysis we get adjusted odds ratio which is more precise estimation of relationship between explanatory and response variable. The adjusted odds ratio is determined by conducting logistic regression analysis, which incorporates all confounding variables into the model. The impact of each explanatory variable on the outcome can be evaluated and understood in light of adjusted odds ratio (Dobson and Barnett, 2018).

3.4.4 Level of Significance and P-value

Level of Significance

The Level of Significance, denoted by α, refers to the threshold probability that is used to determine whether the results of a statistical investigation are statistically significant of the hypothesis test result. It represents the probability of incorrectly rejecting the null hypothesis. In other words, we will say that it refers the probability of type I error. Mathematically the level of significance can be defined as:

$$\alpha = P(\text{Type I error}) = P(\text{Reject } \mathbf{H}_0 \mid \mathbf{H}_0 \text{ is true})$$

α is chosen by the researcher before conducting the hypothesis test. Generally, the value of α is set as 0.05 or 0.10. $\alpha = 0.05$ means that there is about a 5% chance of rejecting a true null hypothesis. α is assigned as 10% for bivariate analysis and adjusted to 5% for the regression model in this study.

P-value

The **P-value** expresses the probability that observed discrepancies happened just by mere chance. The likelihood of receiving a test statistic value that is equally extreme or more extreme than the one calculated under the null hypothesis is known as the p-value. The P-value is derive from the observe data, based on the chi square distribution in the bivariate analysis and the logistic distribution in the the regression analysis. To evaluate whether the two variables are associate or not , p value is compared to the pre-chosen level of significance α. If the p-value is less than the pre-chosen level of significance, α, then the null hypothesis (H_0) is rejected, indicating a significant association between the two variables. On the other hand, if the p-value is greater than the pre-chosen level of significance, α, then the null hypothesis (H_0) is not rejected, suggesting no significant association between the two variables.

3.5 ROC Curve

A Goodness of fit measures how correctly a statistical model represents the observed data. A ROC (Receiver Operating Characteristic) evaluate the effectiveness of binary classifiers, which model are based on binary outcome (e.g., yes/no, true/false) (Fawcett, 2006). This curve displays how the change of the true positive rate (sensitivity) over the different value of the false positive rate (1-specificity). True positive rate, TPR means the the proportion of actual positives that are correctly identified by the model. And, the false positive rate, FPR indicates the he proportion of actual negatives that are incorrectly classified as positives by the model. AUC (area under the curve) of ROC helps to assess the model performance. A value 1 indicates flawless prediction. And the value of AUC is 0.5 is considered a random guess (Hanley and McNeil, 1982). To evaluate the adequacy of the model's fit, the ROC curve was used in the study.

3.6 Statistical Software

STATA and **R** are used for statistical analysis in this study. The version of STATA MP 17.0, and the version of R 4.4.2 used for data analysis, and modeing in this research.

Chapter 4

Resutls and Analysis

4.1 Introduction

The analysis and finding in this study will be presented in this chapter which were from using the methodologies described in the previous chapter. Initially, the exploratory analysis has been conducted which include the frequency and percentage distribution of the characteristics of each independent and dependent variables. Then conduct the bivariate analysis. At first, find the frequency and percentage distribution of the independent variables across each levels of dependent variable. Also, to assess the association between independent and dependent variable, a conduct chi-square test (χ^2) . Finally, a multilevel logistic regression models have been fited.

4.2 Univariate Analysis

4.2.1 Children Attributes

Table 4.1: Univariate Table of children Attributes

Variable	**Percentage (%)**	**Frequency**
Child Survival		
Dead	3.01%	266
Alive	96.99%	8573
Sex of child		
Male	51.21%	4526
Female	48.79%	4313
Birth order		
1st birth	38.15%	3372

Table 4.1 continued from previous page

Variable	Percentage (%)	Frequency
2nd birth	34.07%	3012
3rd or more	27.77%	2455
Birth Weight		
Low birth weight	14.88%	513
Normal birth weight	78.08%	2695
Over birth weight	7.04%	243
Breastfeeding		
Yes	55.66%	4920
No	44.34%	3919
Preceding birth interval		
Short	11.10%	604
Normal	88.90%	4840
Mode of delivery		
Cesarean section	44.09%	2378
Vaginal	55.90%	3016
Place of delivery		
Home	36.25%	1952
Public Hospital	17.08%	920
Private Hospital	46.67%	2514
Birth attendant during delivery		
Trained	61.51%	3326
untrained	38.48%	2080

The table 4.1 shows that approximately 97% of children are alive while only 3% are dead. About 51.2% children are male and 48.8% children are female. The distribution of birth order is roughly equal. 78% children have normal birth weight, and 14.9% children have low birth weight while 7% children have over birth weight in this study. About 55% children are breastfed. A small proportion of Preceding birth interval is short (less than 24 month) . 44% children involves a C-section delivery method while 56% children doesn't involve a C-section delivery method. In this study, 36.25% deliveries occurred at home, 46.67% deliveries occurred at the private hospital and 17.08% deliveries occurred at the public hospital. 61.5% deliveries involves trained person while

38.5% deliveries involves untrained person.

4.2.2 Parental Attributes

Table 4.2: Univariate Table of parental Attributes

Variable	Percentage (%)	Frequency
Mother's Education		
No education	6.20%	548
Primary	23.20%	2059
Secondary	53.66%	4743
Higher	16.84%	1489
Father's Education		
No education	15.94%	1387
Primary	30.74%	2675
Secondary	34.67%	3017
Higher	18.64%	1622
Mother's Occupation		
Working	30.32%	2680
Not working	69.68%	6159
Mother age at child birth		
<20 year	21.90%	1727
$20-30$ year	62.68%	4947
>30 year	15.44%	1219
Antenatal Care (ANC)		
<4	75.78%	3885
$>=4$	24.28%	1242
Media exposure		
Yes	45.20%	3995
No	54.80%	4844

Table 4.2 presents a snapshot of the parental attributes of children. The highest percentage (53.6%) of mothers' educational qualification is secondary, and 6.02% of mothers have no education with the lowest percentage. Similarly, the highest percentage (34.7%) of fathers' educational

qualification is secondary, and the lowest percentage (15.9%) of fathers have no educational qualification. Approximately 30% of mothers are involve in work, while 70% are not are involve in work. Most of the children (62.68%) were born of his mother aged $20 - 29$ years. A fewer percentage of children (15.44%) were born to mothers aged $30 - 49$ years. 75% mother received antenatal care less than 4 times during pregnancy. Meanwhile, 25% women received antenatal care more than four times during pregnancy. About 45% mothers were exposed by media while 55% mothers were not exposed by media in this study.

4.2.3 Household Attributes

Table 4.3: Univariate Table of household Attributes

Variable	**Percentage (%)**	**Frequency**
Division		
Barishal	6.45%	570
Chattogram	21.79%	1926
Dhaka	25.07%	2216
Khulna	10.04%	887
Mymensingh	8.49%	750
Rajshahi	10.25%	906
Rangpur	10.89%	962
Sylhet	7.03%	621
Residence		
Urban	27.00%	2386
Rural	73.00%	6452
Wealth index		
Poorest	20.92%	1849
Poor	20.79%	1837
Middle	20.59%	1820
Rich	19.25%	1701
Richest	18.45%	1630

Table 4.3 visually presents the distribution of division, residence, and wealth index. The distribution of wealth index is almost equal. 73% respondents were from rural areas and the remaining

27% respondents from urban areas.

4.3 Bivariate analysis

In this section, we first try to explore how the prevalence of under five child mortality varies different factors levels. Also, report the p-value derived from the chi-square independence test (χ^2).

4.3.1 Children Attributes

Table 4.4: Bivariate Table of children Attributes

Characteristics	**Child Survival**		**P-Value**
	Dead N(%)	**Alive N(%)**	
Sex of child			
Male	149(3.28%)	4378(96.72%)	0.224
Female	118(2.72%)	4195(97.27%)	
Birth order			
First	96(2.86%)	3275(97.14%)	0.054
Second	77(2.57%)	2934(97.43%)	
3rd or 3rd+	92(3.76%)	2362(96.24%)	
Birth Weight			
Low birth weight	30(5.85%)	483(94.14%)	<0.001
Normal birth weight	31(1.15%)	2664(98.84%)	
Over birth weight	72(4.49%)	3379(95.51%)	
Breastfeeding			
Yes	93(1.89%)	4827(98.11%)	<0.001
No	113(4.42%)	3746(95.58%)	
Preceding birth interval			
Short	29(4.89%)	575(95.11%)	0.019
Normal	139(2.86%)	4702(97.13%)	
Mode of delivery			
Cesarean section	27(1.89%)	2351(98.11%)	<0.001
Vaginal	118(4.42%)	2898(95.58%)	

Table 4.4 continued from previous page

Characteristics	Child Survival		P-Value
	Alive N(%)	**Dead** N(%)	
Place of delivery			
Home	72(3.71%)	1880(96.29%)	<0.001
Public Hospital	33(3.60%)	886(96.40%)	
Private Hospital	38(1.54%)	2475(98.46%)	
Birth attendant during delivery			
Trained	67(2.02%)	3257(97.97%)	<0.001
Untrained	78(3.76%)	2002(96.24%)	

Table 4.4 show the prevalence of under five child mortality across the different levels of the various children attributes. The percentage of under five child mortality differs slightly between male and female children. The under five child mortality of male and female is 3.28% and 2.72%, respectively. The highest percentage of child mortality (3.76%) is observed among children whose birth order three or higher while the lowest percentage is 2.57% among second born children. The highest under-five child mortality occurred among low birth weight newborns while the lowest under-five child mortality occurred among normal birth weight newborns. 1.89% of breastfed newborn babies died within five years, compared to 4.42% babies died within five years who are not breastfed. In our study, we demonstrated that child mortality is higher (4.89%) with children born short preceding birth interval compare to normal preceding birth interval (2.86%). We observed that child mortality decreases when the mode of delivery is Caesarean section (1.89%%) compared to vaginal delivery (4.42%). The place of delivary variable divided into three categories. The mortality percentage is lowest among those delivered in private hospitals and highest among those delivered at home. The percentage of mortality for home deliveries, public hospital deliveries, and private hospital deliveries was 3.71%, 3.60%, and 1.54%, respectively. The mortality rate is lower when a trained person is involved in the delivery compared to when an untrained person is involved.

4.3.2 Parental Attributes

Table 4.5: Bivariate Table of parental Attributes

Characteristics	Child Survival		P-Value
	Dead N(%)	Alive N(%)	
Mother's Education			
No education	18(3.22%)	531(96.78%)	
Primary	71(3.47%)	1988(96.54%)	0.155
Secondary	149(3.12%)	4595(96.88%)	
Higher	29(1.97%)	1459(98.02%)	
Father's Education			
No education	57(4.11%)	1330(95.89%)	
Primary	91(3.42%)	2584(96.58%)	0.001
Secondary	90(2.97%)	2927(97.02%)	
Higher	23(1.40%)	1600(98.60%)	
Mother's Occupation			
Working	171(2.78%)	5988(97.22%)	0.101
Not working	95(3.55%)	2584(96.45%)	
Mother age at child birth			
<20 year	55(3.19%)	1672(96.81%)	0.886
$20-30$ year	144(2.92%)	4803(97.08%)	
>30 year	36(3.02%)	1182(96.98%)	
Antenatal Care (ANC)			
<4	82(2.12%)	3803(97.88%)	0.749
$>=4$	24(1.95%)	1217(96.05%)	
Media exposure			
Yes	127(2.62%)	4717(97.38%)	0.048
No	139(3.49%)	3856(96.51%)	

From the bivariate table of parental attributes 4.5 , we observe that the percentage of under-five child morality differs slightly according to the mother's education. The percentage of child mortality is 3.22%, 3.47%, 3.12%, and 1.97% among children whose mother's have no education, primary, secondary, and higher education, respectively. The percentage of mortality decreases with improved mother's educational level, and a similar trend is observed for the father's educational level. The percentage of child mortality is 4.11%, 3.42%, 2.97%, and 1.40% among children whose father's have no education, primary, secondary, and higher education, respectively. A interesting finding from table 4.5 is that the child mortality is lower among working women (2.78%) compare to non-working women. The ideal age for a mother to take a child is between 20 and 30 years. But there are no remarkable change in rate or child mortality across mother age at child birth. The distribution of child mortality was 3.19%, 2.92%, and 3.02% among mothers aged at child birth less than 20, 20–30, and over 30, respectively. The percentage of child mortality is lower among mothers who had four or more ANC visits (1.95%) compared to those who had fewer than four visits (2.12%). Child mortality rate 2.62% among media exposed mother compare to child mortality rate 3.49% among non-media exposed mother.

4.3.3 Household Attributes

Table 4.6 presents a snapshot of the prevalence of the under-five child mortality of the household attributes of children. The highest prevalence of child mortality occurred in Sylhet (4.45%) and the lowest prevalence occurred in khulna (2.14%). And other divisions don't very remarkably. The rate of child morality 2.95% in the urban areas, and this rate increases to (3.03%) the rural areas. Child mortality tends to decrease with improvements in economic conditions. The prevalence of child mortality is 4.41% among poorest family and 1.97% among rich family.

Table 4.6: Bivariate Table of Household Attributes

Characteristics	**Child Survival**		**P-Value**
	Dead N(%)	**Alive N(%)**	
Division			
Barishal	18(3.21%)	552(96.80%)	
Chattogram	59(3.08%)	1866(96.92%)	
Dhaka	56(2.52%)	2160(97.48%)	0.213
Khulna	18(2.14%)	868(97.86%)	
Mymensingh	23(3.08%)	727(96.92%)	
Rajshahi	26(2.83%)	881(97.17%)	
Rangpur	38(3.92%)	925(96.08%)	
Sylhet	27(4.45%)	593(95.55%)	

Table 4.6 continued from previous page

Characteristics	Child Survival		P-Value
	Alive	**Dead**	
	N(%)	**N(%)**	
Residence			
Urban	70(2.95%)	2316(97.05%)	0.861
Rural	196(3.03%)	6257(96.97%)	
Wealth index			
Poorest	81(4.41%)	1768(95.59%)	
Poor	66(3.58%)	1772(96.42%)	<0.001
Middle	51(2.78%)	1769(97.22%)	
Rich	33(1.97%)	1668(98.03%)	
Richest	35(2.14%)	1596(97.86%)	

4.4 Results from Multilevel Logistic Regression Model

The model then estimate the coefficents for each predictor variable, which can be used to make predictions about the probability of the outcome variable being a "success" given a set of predictor variable values.

Table 4.7: Results of multilevel logistic regression analysis of under-five child Mortality

Variables	Adj odds ratio	P-Value	[95% Conf. Interval]
Division			
Mymensingh	(Ref.)		
Barishal	0.444	0.275	(0.104,1.904)
Chittagong	0.116	0.004	(0.027,0.506)
Dhaka	0.168	0.014	(0.040,0.699)
Khulna	0.451	0.259	(0.113,1.798)
Rajshahi	0.359	0.190	(0.078,1.662)
Rangpur	0.968	0.962	(0.264,3.551)
Sylhet	0.233	0.049	(0.051,0.997)
Wealth Index			
Poorest	(Ref.)		
Poorer	0.879	0.450	(0.630,1.227)
Middle	0.713	0.062	(0.500,1.016)
Richer	0.511	0.193	(0.257,1.351)
Richest	0.567	0.236	(0.401,15.721)
Mother's Occupation			
Not working	(Ref.)		
working	0.744	0.498	(0.317,1.749)
Mother's education			
No education	(Ref.)		
primary	0.975	0.924	(0.591,1.612)
Secondary	0.899	0.663	(0.559,1.447)
Higher	0.957	0.031	(0.302,0.945)

Table 4.7 continued from previous page

Variables	Adj odds ratio	P-Value	[95% Conf. Interval]
Father's education			
No education	(Ref.)		
primary	4.552	0.027	(1.192,17.386)
Secondary	2.020	0.323	(0.500,8.157)
Higher	0.957	0.962	(0.158,5.811)
Media exposure			
No	(Ref.)		
Yes	0.804	0.962	(0.633,1.022)
Sex of child			
female	(Ref.)		
male	2.117	0.045	(1.016,4.414)
Birth order			
First	(Ref.)		
Second	0.845	0.270	(0.628, 1.138)
3rd or 3rd+	1.264	0.106	(0.951,1.681)
Birth weight			
Low	(Ref.)		
Normal	0.113	<0.001	(0.051,0.250)
Over	0.428	0.153	(0.134,1.368)
Breastfeeding			
No	(Ref.)		
Yes	0.052	<0.001	(0.023,0.113)
Preceding birth interval			
Short	(Ref.)		
Normal	0.597	0.309	(0.221,1.614)
Mode of delivery			
Vaginal	(Ref.)		
Caesarean section	0.307	0.009	(0.125,0.746)

Table 4.7 continued from previous page

Variables	Adj odds ratio	P-Value	[95% Conf. Interval]
Place of delivery			
Home	(Ref.)		
Public Hospital	1.080	0.905	(0.306,3.817)
Private Hospital	1.071	0.880	(0.284,4.047)
Birth attendant during delivery			
Trained	(Ref.)		
untrained	2.288	0.089	(0.882,5.933)
Intercept	0.358	0.473	(0.022,5.885)
Random Cluster Effect	0.4834		(0.304,0.766)

Table 4.7 provides valuable insights into the relationships between different risk factors and under-five child mortality. Each factor's reference category is indicate odds ratio, OR= 1. Below are comprehensive for each variable and its respective categories.

Mymensingh is used as a reference category to examine regional disparities in child mortality. Barishal experienced a statistically insignificant decrease of 46% in odds (OR = 0.44, 95% CI: 0.10,1.90, p value = 0.27). Chittagong showed a statistically significant 88% reduction in odds (OR = 0.12, 95% CI: 0.03,0.51, p value = 0.004). Dhaka recorded a statistically significant decrease of 83% odds in child morality (OR = 0.17, 95% CI: 0.04,0.70, p value = 0.014). Khulna, Rajshahi, and Rangpur experienced a statistically insignificant reduction in child mortality by 55%, 64%, and 3%, respectively. However, Sylhet recorded a statistically significant decrease of 77% odds in child morality compare to Mymensingh (OR = 0.23, 95% CI: 0.05,0.99, p value = 0.049). The wealth index categories were assessed relative to the poorest group. The child mortality within poorer group decreased by 12% compare to poorest group (OR = 0.88 95% CI: 0.63,1.22, p value = 0.45). The middle and richer socioeconomic category exhibits a statistical insignificant decreases of 29%, and 49%, respectively, in the likelihood of under-five mortality. Also, a statistical insignificantly reduction of child mortality within the richest group by 43% compared to the poorest group. If mothers are engaged in employment, there is a statistically insignificant 26% reduction in the likelihood of their babies morality chance compared to babies whose mothers are not employed.

The reference category of Mother's education is No education. Mother's education has a positively significant impact on child mortality. The probability of child mortality among babies whose mother's have primary, secondary, and higher education is 2.5%, 10%, and 5% lower, respectively, compare to babies whose mother's have no education. similarly, The reference category of Fother's education is No education. Child mortality is 355% and 102% higher among children whose fathers have primary and secondary education, respectively. But, 2.5% lower child mortality among children whose father's education is higher with statistically insignificant. Mother who exposed by media, such as watching tv, listening radio, or reading newspaper had a 20% reduction in child mortality (OR = 0.80, 95% CI: 0.63,1.02, p value = 0.962) compared to mothers who were not exposed to media.

The reference category of sex of child is female. The odds ratio (OR) of male is 2.12, which is greater than 1. So, The probability of getting chance of mortality is higher than female (OR = 2.12, 95% CI: 1.02,4.41, p value = 0.045). Therefore, this factor had a significant effect on child mortality. Second birth odds ratio (OR) is 0.845 meaning the second child has 20% reduction in the likelihood of child mortality compared to first child. However, a third or higher-order child (3+) has a 26% increase in the likelihood of child mortality compared to the first child. This factor had no significant effect on child mortality (p-value: 0.138).

Birth weight has a significant effect on child mortality. The reference category of birth weight variable is low birth weight (<2500 gm). There is an 89% lower likelihood of child mortality among normal birth weight babies compared to low birth weight babies. Additionally,, 52% lower likelihood of child mortality among over birth weight babies compared to low birth weight babies. Breastfeeding is another important variable that has a significant impact on child mortality. There is 95% lower likelihood of the child mortality among breastfed children compared to non-breastfed children(OR = 0.052, 95% CI: 0.02,0.22, p value = <0.001).

Preceding birth interval has no significant impact on under-five child mortality. The reference category of Preceding birth interval is short Preceding birth interval (<24months). A normal preceding birth interval (24 months) reduces the odds of under-five child mortality by 39% (OR = 0.59, 95% CI: 0.22,1.61, p value = 0.31).

Caesarean section (OR = 0.30, 95% CI: 0.12,0.75, p value = 0.009) is associated with a statistically significant 70% reduction in the likelihood of child mortality compared to vaginal delivery. The reference was place od delivery is home. Public hospital exhibited a statistically insignificant

8% increase in odds of under-five child mortality and private hospital increase 7% risk of child mortality. Birth attendant during delivery is another variable. In this case, Trained person include in the delivery is the reference category this variable. Deliveries attended by untrained persons are a statistically insignificant associated with a 128% increase in the likelihood of child mortality (OR = 2.28, 95% CI: 0.88,5.93, p value = 0.089).

Given the design of the survey, the cluster numbers serve as the chosen level for the multilevel model. The presence of a random cluster effect significantly impacts our fitted model, as evidenced by the confidence interval of its odds ratio not encompassing the value 1.

As a result, we see that Division, Wealth Index, Mother's education, Father's education, Sex of child, Birth weight, Breastfeeding, and Mode of delivery are found to be significantly associated with under-five mortality. These factors highlight the need for targeted interventions in maternal and child health programs.

Intraclass Correlation Coefficient (ICC)

The ICC of this model is 0.122 and 95% CI is (0.092 ,0.207). The **ICC** indicates that approximately 12.2% of the variation in the outcome variable can be attributed to cluster variation.

4.5 Goodness of Fit (GOF):

The assessment of the fitted model's goodness of fit is confirmed through the ROC curve analysis. Below are the ROC findings for our fitted model, which provide insights into its performance and discriminatory power-

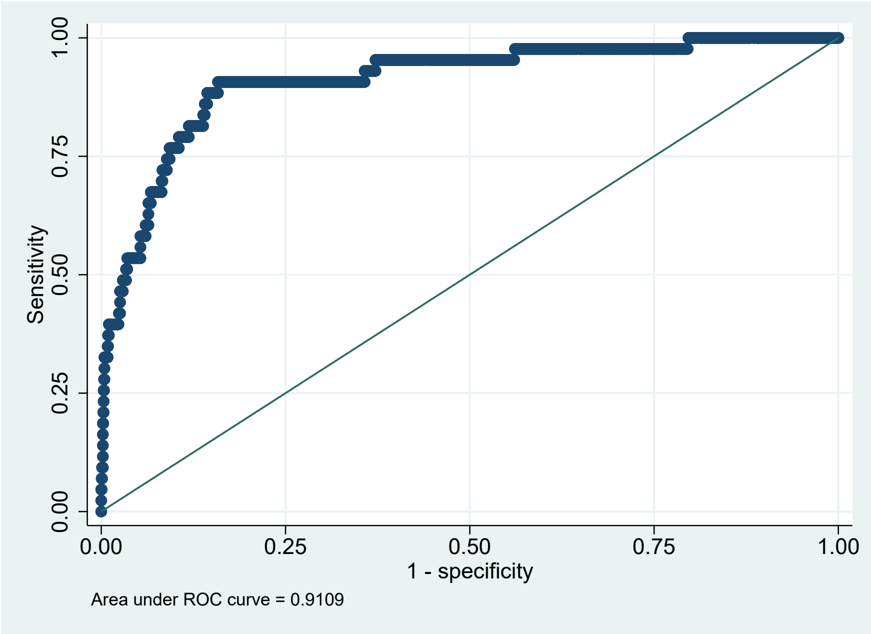

Figure 4.1: **Receiver Operating Characteristic (ROC) curve: result from mixed effect model**

The ROC curve 4.1 corresponding to mixed effect model displays an AUC value of 0.9109, indicating a high level of discrimination between the two classes. This suggests that the classifier can effectively distinguish between positive and negative samples, achieving accurate classifications approximately 91.09% of the time.

Chapter 5

Discussion and Conclusions

5.1 Discussion

The focus of this study is to examine the prevalence and associated factors of mortality among children in Bangladesh aged 0-5 years. Also focus of the study is to create a predictive model that would help Bangladesh identify risk factors for under-five child mortality. A bi-variate analysis has given the initial idea about the associated factors affecting the under-five child mortality. The research utilizes a multilevel binary logistic model that considers various factors such as division, mother's education, father's education, wealth index, birth order, and birth weight etc.

Paternal and matarnal education is necessary to significantly lower Bangladesh's overall rate of under-five mortality in their study (Arntzen et al., 1993). The findings from our analysis indicate a significant association between Mother's education and under-five child mortality. Specifically, individuals whose mothers have higher levels of education demonstrate lower odds of child mortality compared to those whose mothers have no education. The mortality rate is 3.22% among children of mothers with no education, while it is only 1.97% among children of highly educated mothers This finding resonates with prior research, exemplified by (Wu, 2022), whose investigation underscored that an additional year of maternal schooling can reduce under-five child mortality by approximately 20%. In addition, father education has a significant association with child mortality. Similarly, an increase in father's education is associated with a decrease in under-five child mortality. Our research echoes the findings of (Balaj et al., 2021) indicating that 1·57% mortality reduction for a one year additional paternal education.

(Briend et al., 1988) has been discovered that breastfeeding reduce child mortality. This observation aligns with the results of our study. Our study shows a 95% reduction in mortality due to breastfeeding. Our study depicts that Caesarean section reduces under-five child mortality by 70%. A similar result was reported in a previous study (Razzaque et al., 2024), which showed that Caesarean section deliveries reduced child mortality by 51%. This study found that normal birth weight reduces child mortality compared to low birth weight, which is consistent with the research conducted by (Baqui et al., 2008).

This study further shows that male children are more likely to experience child mortality than female children to child mortality. which is similar to the study conducted by (Hossain et al., 2015). This variable is significant in both my study and theirs. Child mortality exhibits regional disparities, varying significantly across different divisions. A similar result was shown by (Khan and Awan, 2017). This study shows that Sylhet division is the most vulnerable, whereas our study shows that Mymensingh division is the most vulnerable.

5.2 Conclusion

This study investigates the prevalence as well as variables that impact under-five child mortality. Under-five child mortality were significantly linked to mother's education, father's education, sex of children, birth weight, breastfeeding, mode of delivery, and division. Along with other development partners, the government of Bangladesh need to encourage policies and initiatives that, by taking into account factors like mother's education, father's education and regional inequalities in Bangladesh. The government should encourage people to opt for C-section deliveries and to promote breastfeeding. Our study has been done to play small but significant role in achieving Bangladesh goals in Sustainable Development Goal (SDG) 3.2. Bangladesh, like many other countries, aligns its efforts to reduce child mortality with the global Sustainable Development Goal (SDG) 3.2, which aims to end preventable deaths of newborns and children under five years old, with a specific target to reduce neonatal mortality to at least 12 per 1,000 live births and under-five mortality to at least 25 per 1,000 live births by 2030. We anticipate that the study's findings contribute to realizing the national goal of achieving the lowest child mortality rate by 2030, aligning with the broader vision for 2041.

5.3 Limitations

The large and nationally representative data set of this study is one of its most vital points. As a result, the findings can be applied nationally. As a result, we have offered a nationwide assessment of Bangladesh's under-five child mortality. This enables the findings to be utilized on a national level. This study has certain drawbacks in addition to its benefits. Secondary data were employed in this investigation. Although it is secondary data, it may be possible to identify some significant variables that are also associated with under-five child mortality, but we were not able to incorporate them into this study. The variables present in the dataset limited our ability to account for every possible factor that could be correlated with under-five child mortality. Again, the cross-sectional and nationally representative nature of the data set limits its ability to demonstrate a causal relationship between the associated characteristics and the prevalence of under-five child mortality.

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